

EXPERIMENT 7B

DATA VISUALIZATION USING TABLEAU – Sales Dataset

Aim:

To connect, analyze, visualize, and create interactive dashboards using Tableau for effective business intelligence and data analytics.

Sample Dataset:

Sales Dataset

A synthetic sales dataset containing 80 records and 12 columns, created to analyze sales performance across regions, cities, product categories, customer segments, and payment modes using Tableau.

Algorithm 1:

Connecting Data Source

Step 1 Open Tableau Desktop

Step 2 Select Connect

Step 3 Choose Excel File

Step 4 Browse Dataset

Step 5 Open Dataset

Step 6 Verify Imported Data

Algorithm 2:

Step 1: Open a new worksheet.

Step 2: Drag Region → Rows.

Step 3: Drag Sales → Columns.

Step 4: Select Bar Chart visualization.

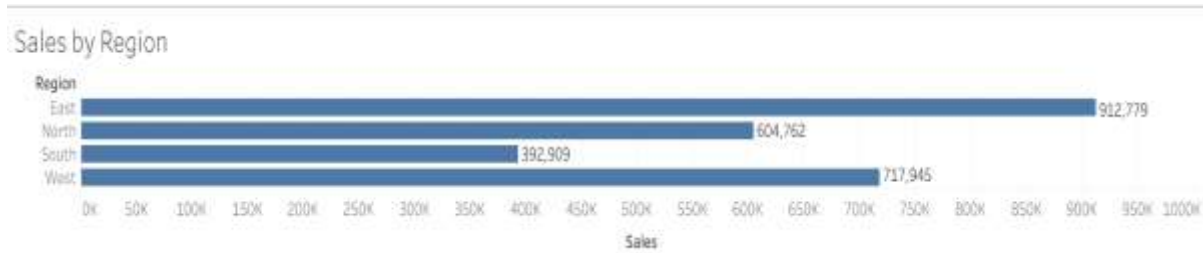
Step 5: Apply data labels.

Step 6: Analyze sales across different regions.

Output:

Sales by Region bar chart.

The bar chart compares total sales across North, South, East, and West regions.



Algorithm 3:

Step 1: Create a new worksheet.

Step 2: Drag Region → Rows.

Step 3: Drag Sales → Columns.

Step 4: Select Bar Chart.

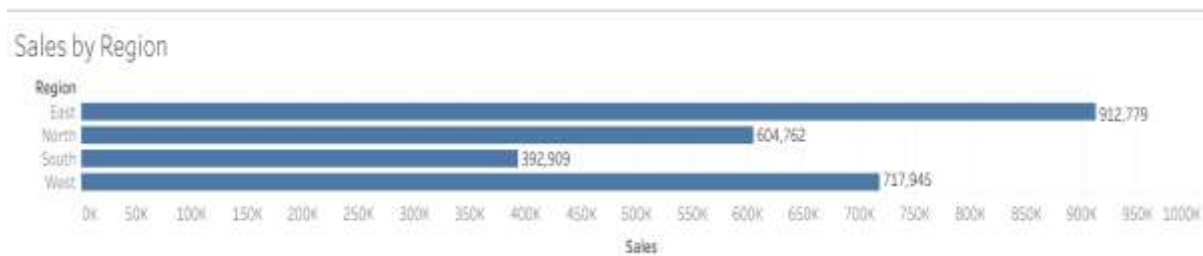
Step 5: Apply data labels.

Step 6: Format the visualization.

Output:

Sales by Region bar chart.

The bar chart compares total sales across North, South, East, and West regions



Algorithm 4:

Step 1: Create a new worksheet.

Step 2: Select Pie Chart.

Step 3: Drag Category → Color.

Step 4: Drag Sales → Angle.

Step 5: Drag Sales → Label.

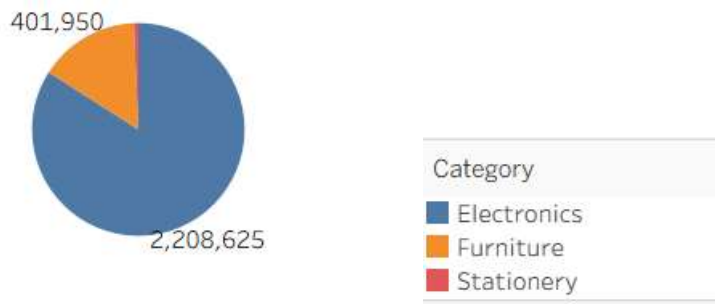
Step 6: Analyze the contribution of each category.

Output:

Category-wise Sales pie chart.

The pie chart represents the contribution of Electronics, Furniture, and Stationery to total sales.

Category-wise Sales



Algorithm 5:

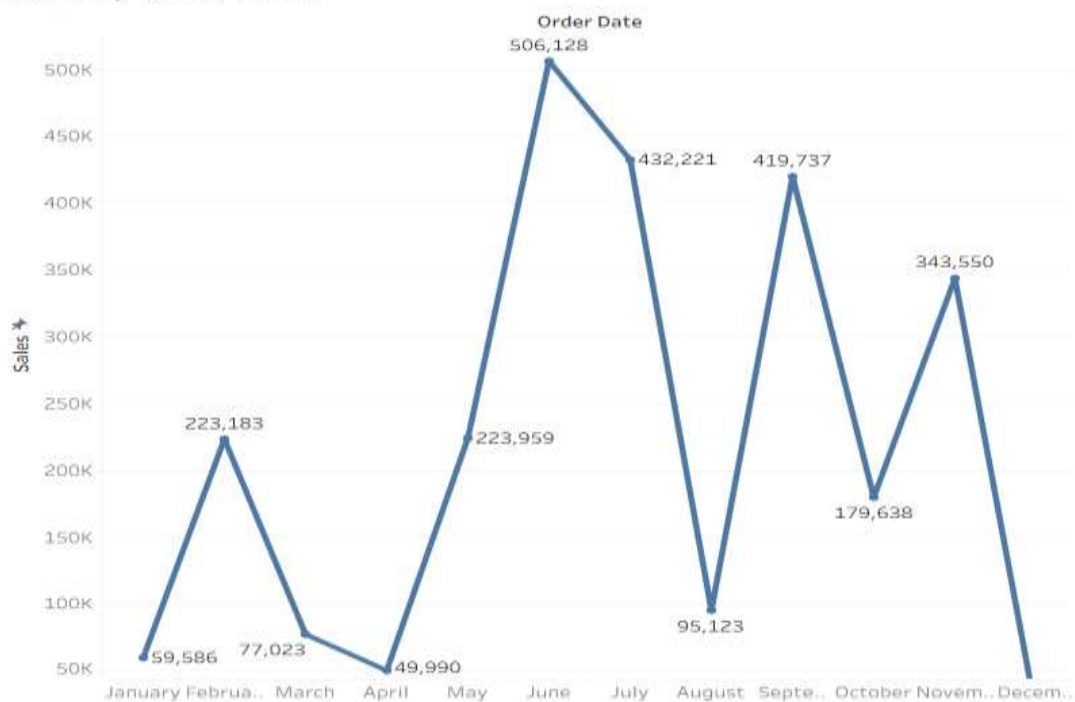
- Step 1: Create a new worksheet.
- Step 2: Drag Order_Date → Columns.
- Step 3: Set the date level to Month.
- Step 4: Drag Sales → Rows.
- Step 5: Select Line Chart.
- Step 6: Apply appropriate formatting.

Output:

Monthly Sales Trend line chart.

The line chart shows the variation in sales over different months.

Monthly Sales Trend



Algorithm 6:

Creating Calculated Field

Objective

To calculate the Profit Margin from Sales and Profit.

Profit Margin = $\text{SUM}(\text{Profit}) / \text{SUM}(\text{Sales})$ Procedure

Step 1: Select Analysis → Create Calculated Field.

Step 2: Enter the name Profit Margin.

Step 3: Enter the formula:

$\text{SUM}([\text{Profit}]) / \text{SUM}([\text{Sales}])$

Step 4: Click OK.

Step 5: Use Profit Margin in the visualization.

Step 6: Format the value as a percentage.

Output:

Profit Margin by Category visualization.

The calculated field helps compare the profitability of different product categories.



Category	
Electronics	15.49%
Furniture	14.79%

Algorithm 7:

Step 1: Select the required worksheet.

Step 2: Drag Region → Filters.

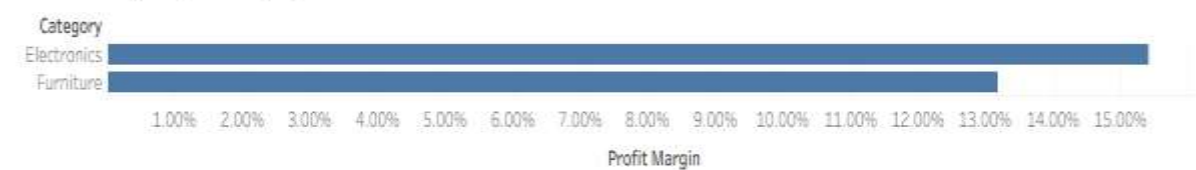
Step 3: Select the required region.

Step 4: Apply the filter.

Step 5: Observe the updated visualization.

Output:

Profit Margin by Category



Sales Threshold

☐ (All)

☐ False

☒ True

Region

☐ (All)

☐ East

☒ North

☐ South

☐ West

Algorithm 8:

To analyze sales records based on a user-defined minimum sales value.

Step 1: Select Analysis → Create Parameter.

Step 2: Name the parameter Minimum Sales.

Step 3: Set the data type and allowable range.

Step 4: Create the calculated field Sales Threshold.

Step 5: Use the formula:

$[Sales] \geq [Minimum\ Sales]$

Step 6: Apply Sales Threshold = True as a filter.

Step 7: Display the parameter control.

Step 8: Change the minimum sales value and observe the updated visualization.

Output:

Interactive Minimum Sales Parameter.

The parameter allows users to dynamically analyze sales records above a selected sales threshold

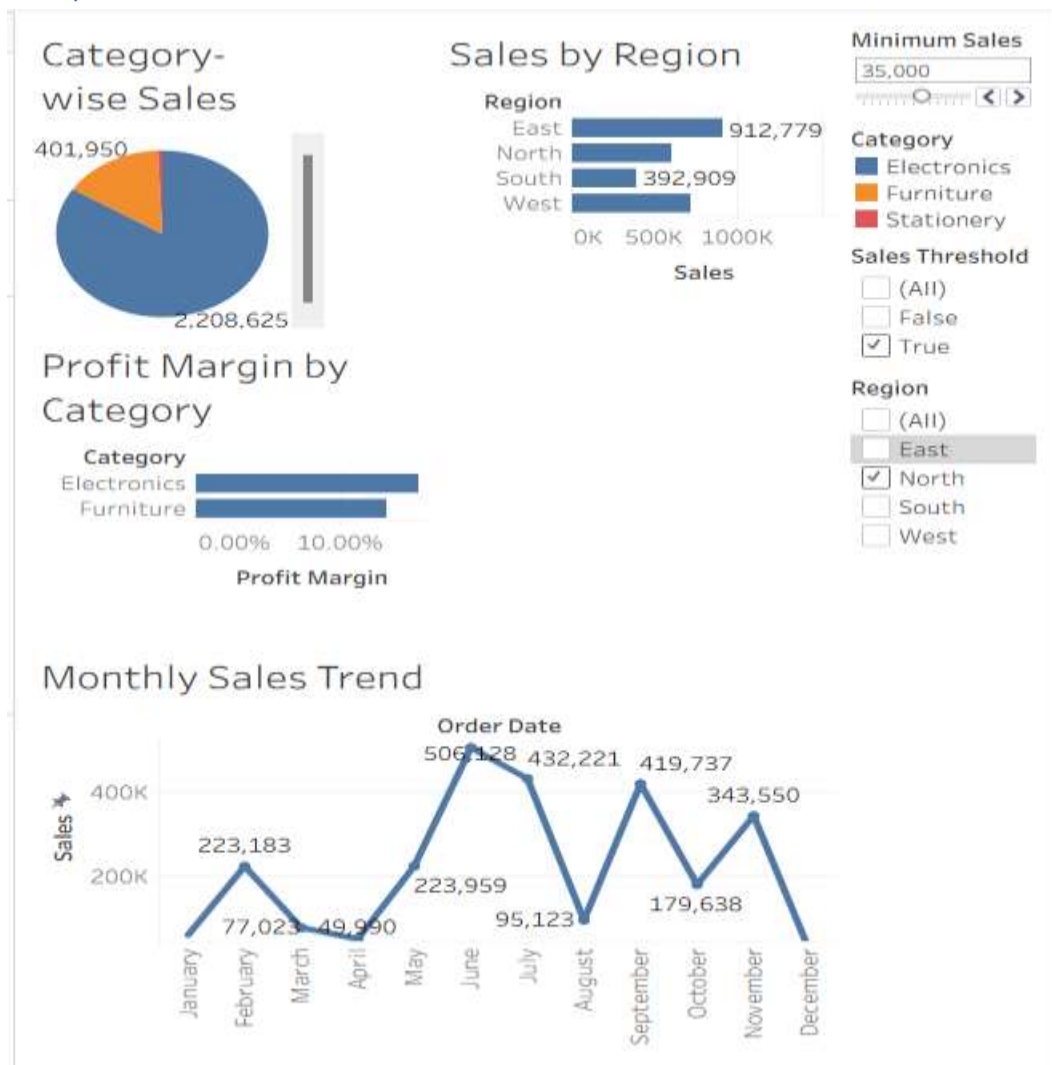
Minimum Sales

35,000

Algorithm 9:

- Step 1: Create the required worksheets.
- Step 2: Select Dashboard → New Dashboard.
- Step 3: Drag the worksheets onto the dashboard.
- Step 4: Arrange the visualizations appropriately.
- Step 5: Add the Region Filter and Minimum Sales Parameter.
- Step 6: Format the dashboard.
- Step 7: Save the dashboard.

Output:



Result:

Thus, Tableau was used successfully to connect datasets, create worksheets, apply filters and parameters, generate visualizations, build dashboards, and create stories for business intelligence and analytical decision-making.